FIITJEE

JEE(Main), 2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST - IV
(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	31.	**C**	61.	**D**
2.	**C**	32.	**D**	62.	**D**
3.	**A**	33.	**C**	63.	**A**
4.	**A**	34.	**A**	64.	**C**
5.	**A**	35.	**A**	65.	**C**
6.	**A**	36.	**C**	66.	**B**
7.	**D**	37.	**A**	67.	**C**
8.	**B**	38.	**B**	68.	**C**
9.	**C**	39.	**A**	69.	**C**
10.	**C**	40.	**A**	70.	**B**
11.	**A**	41.	**C**	71.	**D**
12.	**A**	42.	**A**	72.	**B**
13.	**A**	43.	**D**	73.	**B**
14.	**B**	44.	**B**	74.	**D**
15.	**A**	45.	**C**	75.	**C**
16.	**C**	46.	**C**	76.	**B**
17.	**A**	47.	**C**	77.	**B**
18.	**B**	48.	**D**	78.	**B**
19.	**B**	49.	**C**	79.	**B**
20.	**D**	50.	**C**	80.	**B**
21.	**A**	51.	**D**	81.	**D**
22.	**B**	52.	**A**	82.	**A**
23.	**C**	53.	**A**	83.	**D**
24.	**B**	54.	**D**	84.	**C**
25.	**D**	55.	**A**	85.	**D**
26.	**B**	56.	**C**	86.	**A**
27.	**B**	57.	**A**	87.	**C**
28.	**A**	58.	**C**	88.	**C**
29.	**C**	59.	**B**	89.	**C**
30.	**B**	60.	**B**	90.	**A**

Physics

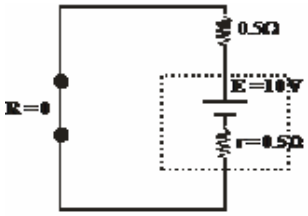
PART – I

SECTION – A

1. D

Sol. For maximum power transfer in external circuit R = 0;

Now circuit will become



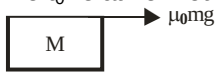
Power transfer in external resistance 0.5 Ω is same as power transfer in external circuit

$$=\left(\frac{10}{0.5+0.5}\right)^2\times 0.5=50\,\text{watt}$$

So, correct answer (D)

2. C

Sol. Let at time t_0 relative motion will occur



$$\mu_0 mg = Ma \quad \text{..... (1)}$$



$$10\,t_0-\mu_0 mg=ma \quad \text{..... (2)}$$

From (1) and (2) ;

$$t_0=\mu_0 m+\frac{\mu_0 m^2}{M}$$

3. A

Sol. $$Mg-T=M\frac{dv}{dt} \quad \text{..... (1)}$$

$$T+\mu v_e-(M_0-\mu t)g=(M_0-\mu t)\frac{dV}{dt} \quad \text{..... (2)}$$

From (1) and (2), we get

$$Mg+\mu V_e-(M-\mu t)g=(2M-\mu t)\frac{dv}{dt} \quad \text{..... (3)}$$

$$(\mu v_e+\mu gt)=(2M-\mu t)\frac{dv}{dt} \quad \text{..... (4)}$$

$$\frac{dv}{dt}=\frac{\mu(v_e+gt)}{(2M-\mu t)}$$

$$\therefore \quad \frac{dv}{dt}=\frac{\mu(v_e+gt)}{(2M-\mu t)}$$

So, correct answer is (A)

4. A

5. A

Sol. $\frac{l}{K_1A}+\frac{l}{2K_2A}+\frac{l}{K_3A}=\frac{l}{KA}+\frac{l}{2K_A}+\frac{l}{KA}$

$$\frac{5}{2K}=\left(\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right)$$

$$K_{eq}=\frac{5}{2\left(\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right)}$$

6. A

Sol. Voltage across capacitor will lag current by 90° ;

$$\therefore \quad i_C=\frac{200\sqrt{2}}{10}\sin\left(\omega t+\frac{\pi}{4}+\frac{\pi}{2}\right)$$

$$i=20\sqrt{2}\sin\left(\omega t+\frac{3\pi}{4}\right)$$

$\therefore$ Correct answer is (A)

7. D

Sol. 2dQ = dU + 2dW

dQ = dU + dW

dQ = dW

$\Rightarrow$ dU = 0

Process is isothermal.

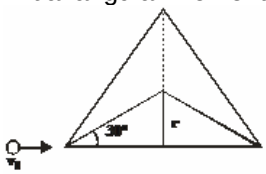
8. B

Sol. F = q v × B

= q v_0i × B (–k) = qv_0 B j

9. C

Sol. Initial angular momentum = final angular momentum



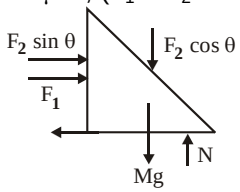
$$L_I=mv_0\frac{l}{2\sqrt{3}}=I_{cm}\omega$$

$$mv_0\frac{l}{2\sqrt{3}}=3m\left(\frac{l}{\sqrt{3}}\right)^2\omega \ ; \ \omega=\frac{v_0}{2\sqrt{3}l}$$

$$y=\frac{l}{2}\tan 30^o=\frac{l}{2\sqrt{3}}$$

10. C

Sol. f = μN ; (F_1 + F_2 sin θ) = μ (mg + F_2 cos θ)



$$\mu = \frac{F_1 + F_2 \sin\theta}{mg + F_2 \cos\theta}$$

11. A

Sol. $W_B = 0$

$W_E = \Delta KE \;\; ; \; W_B + W_E = \Delta KE$

12. A

K.E. = $p^2 / 2m$ and $p = h / \lambda$

13. A

14. B

Sol. Conservation of energy

$$\frac{1}{2}mv_0^2 = mgR + \frac{1}{2}mv^2, v = v_0\cos\theta$$

$$\frac{1}{2}v_0^2 = gR + \frac{v_0^2\cos^2\theta}{2}$$

$$v_0^2\left[1-\cos^2\theta\right] = gR$$

$$v_0 = \frac{\sqrt{2gR}}{\sin\theta}$$

15. A

Sol. N_b reaches maximum when activity of A will be equal to activity of B.

i.e., $\lambda_a N_a = \lambda_b N_b \Rightarrow N_b = \frac{\lambda_a N_a}{\lambda_b}$

16. C

Sol. $f_0 = \frac{c}{\lambda} = \frac{c}{4(l+e)}$ and $\frac{c}{c_0} = \sqrt{\frac{273+\theta}{273}}$

$$\therefore \; f_0' = \frac{c_0}{4(l+e)}\sqrt{1+\frac{\theta}{273}}$$

If temperature is same, first overtone will be $3f_0$

$\therefore$ at 273°C first overtone is $3\sqrt{2}\,f_0$.

17. A

Sol. Net retardation $a = \frac{F_{ext} + \mu_k mg}{m} = 4.5$

If body stop at time t, then

$V = u + at$

$0 = 45 - 4.5t \;\; \Rightarrow t = 10\,sec$

When block stops, F_{ext} will try to bring the block back ward while frictional force will oppose its motion, since block is stationary therefore at this moment, frictional force will be static. Whose maximum value will be $\mu_s mg = 30N$. Since static frictional force is self adjusting therefore it will be 25 N and block will not move after t = 10 sec.

$S = \frac{u^2}{2a} = 225\,m$

18. B

Sol. Time taken by the layer of ice to increase its thickness by $x = x_2 - x_1$

$$t = \frac{\rho L}{2K\theta}(x_2^2 - x_1^2) = \frac{.9\times 80}{2\times .005\times 10}\left[(15)^2 - (10)^2\right]$$

$t = 9\times 10^4 \text{ sec}$; t = 25 Hours

19. B

Sol. At any time

$$Kx - IlB = ma \quad \ldots\ldots \quad (1)$$

$$\frac{q}{C} = Bvl$$

$$I = BlCa \quad \ldots\ldots \quad (2)$$

From (1) and (2)

$$kx = a(B^2l^2C + m)$$

$$a = \left(\frac{K}{B^2l^2C + m}\right)x \quad \ldots\ldots \quad (3)$$

Since motion of connector will be opposite to the displacement from equilibrium. Therefore equation (3) represents S.H.M.

$$\omega = \sqrt{\frac{K}{B^2l^2C + m}}$$

$$V_{max} = X_0\omega$$

$\therefore$ $q_{max} = BlCV_{max}$

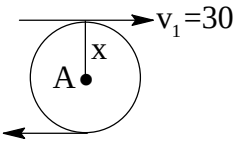
$$q_{max} = BlCX_0\omega$$

$$X_0\sqrt{\frac{KC}{1 + \frac{m}{B^2l^2C}}}$$

20. D

Sol. Let x be the distance of axis of rotation from the top of sphere A (downward). Since angular velocity of the top and bottom point of sphere A will be same, therefore,

$$\omega = \frac{30}{x} = \frac{10}{(2R - x)}$$



$$\frac{30}{x} = \frac{10}{20 - x}$$

$\Rightarrow$ x = 15

$\therefore$ Distance from ground

= 2R + (2R – x)

= 20 + 5 = 25 cm

21. A

Sol. Velocity of image of particle B

$$\overline{v}_B = 5(\hat{i}) + 4(-\hat{i}) = \hat{i}$$

velocity of image of particle A

$$v_A = 10(-\hat{i}) + 4(-\hat{i}) - 10\hat{j}$$

$= 14(-\hat{i}) - 10\,\hat{j}$

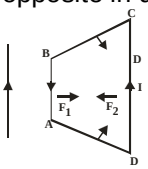
Relative velocity of image

$\overline{V}_{BA} = \overline{V}_B - \overline{V}_A = \hat{i} - \left[14(-\hat{i}) - 10\hat{j}\right] = 15\hat{i} + 10\,\hat{j}$

$|\overline{V}_{BA}| = \sqrt{325}\,\text{cm/sec}$

22. B

Sol. Magnitude of force on AB (F_1) and magnitude of force on CD (F_2) will be equal in magnitude and opposite in direction.



Hence $\overline{F}_1 + \overline{F}_2 = 0$ resultant of $\overline{F}_3 + \overline{F}_4$ will be away from the long straight wire.

23. C

Sol. At any time

$F_{thrust} - F_{friction} = ma$

$\rho a v^2 - \mu mg = ma$

$\rho a(2gh) - \left(\frac{a}{A}\right)(\rho Ah)g = (\rho Ah)a'$

$2\rho agh - \rho agh = \rho Aha'\ ;\ \frac{a}{A}g = a'$

24. B

25. D

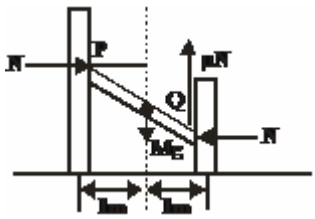
Sol. Drawing F.B.D. of rod PQ

$\Sigma F_x = 0$

$\Sigma F_y = 0 \Rightarrow \mu N = mg$ ..... (1)

$\Sigma\tau$ about centre of mass = 0

$\left[N \times \frac{l}{2}\sin\theta\right][2] = \mu N \times \frac{l}{2}\cos\theta \Rightarrow \tan\theta = \frac{\mu}{2}$



Hence $\sec\theta = \left(\sqrt{\mu^2 + 4}\right)/2$

Thus, $\left(\frac{l/2}{1}\right) = \frac{\sqrt{\mu^2+4}}{2} \Rightarrow l = \sqrt{\mu^2+4}$

$\Rightarrow\ l = \sqrt{0.25+4} = \frac{\sqrt{17}}{2}\,\text{metres}$

26. B

Sol. $e = -\frac{d\phi}{dt}$

$i = \frac{e}{R}$ where R is resistance

$H = i^2 Rt$

The emf will induce from zero to some maximum value and then from maximum to zero.

27. B

Sol. $V_0 = \sqrt{\frac{GM}{3R/2}} = \sqrt{\frac{2GM}{3R}}$

In order to escape to infinity let $(V - V_0)$ is the speed change required. Thus,

$\frac{-GMm}{(3R/2)} + \frac{1}{2}m(v)^2 = 0$

Thus, $v = \sqrt{\frac{4GM}{3R}}$

Thus speed change required = $v - v_0$

$\Delta V = \sqrt{\frac{4GM}{3R}} - \sqrt{\frac{2GM}{3R}}$; $\Delta V = \sqrt{\frac{GM}{3R}}\left(2 - \sqrt{2}\right)$

28. A

Sol. The torque due to liquid in annular space is

$$\left(2\pi R_1 H\right) \times \mu \left(\frac{\omega R_2}{R_2 - R_1}\right) \times R_1$$

where velocity gradient $= \frac{\omega R_2}{R_2 - R_1}$

Also torque due to liquid in between space in bottom of thickness 'h'

$\int_0^{R_1} (x)\left(2\pi x dx\right) \cdot \frac{\omega x}{h} = \frac{\pi \mu \omega R^4}{2h}$

29. C

Sol. $t = \int \frac{dy}{V_y} = \int_0^d \frac{dy}{Ky^{1/2}} = \frac{2\sqrt{d}}{k}$.

30. B

Sol. The volume of liquid should remain unchanged.

Hence, $2 \times 2 \times 2 = 2 \times \frac{1}{2}\left[x + x + \frac{2}{\sqrt{3}}\right] \times 2$

$\therefore \quad x \approx 1.42\,m$

Now, $h = x \sin 60° = 1.23$ m

$\therefore \quad v = \sqrt{2gh} = \sqrt{2 \times 10 \times 1.23} = 4.96\,m/s$

Chemistry

PART – II

SECTION – A

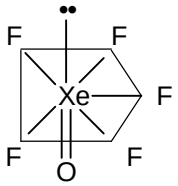
31. C

Sol. Ionic mobilities of hydrated alkali metal ions are as:

$Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$

32. D

Sol.



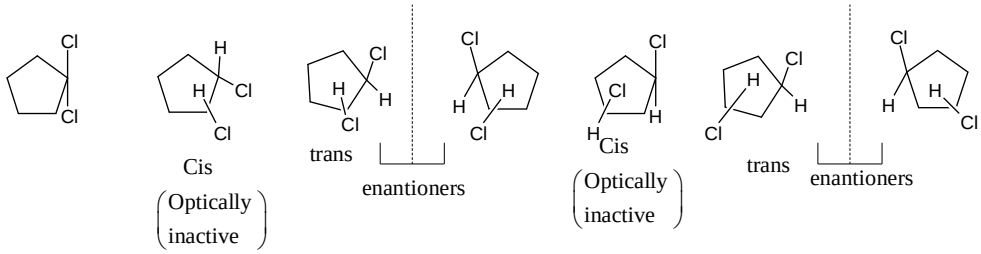
Hybridisation $-sp^3d^3$

Shape – pentagonal monopyramidal

33. C

Sol. Macromolecular colloids like starch, gelatin are generally lyophilic.

34. A

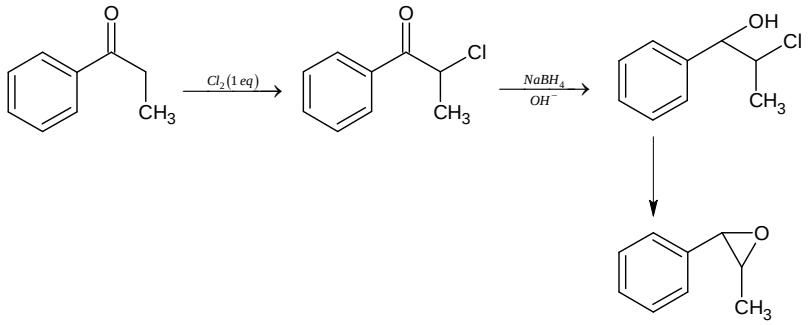


Total isomers = 7

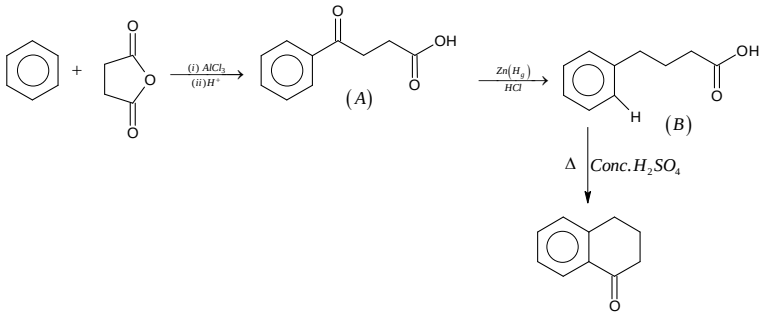
35. A

Sol. Groups to be eliminated should be trans diaxial in chair conformation.

36. C



37. A



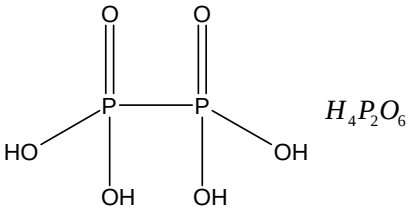
38. B

$$Bi^{3+} \xrightarrow{KI} BiI_3 \xrightarrow[KI]{excess} \left[BiI_4\right]^-$$

(A) (B)

Black ppt yellow brown solution

39. A



$H_4P_2O_6$

40. A

Because of stability of half filled f-subshell configuration.

41. C

$AgBr = x + y$

$$AgBr = \frac{K \times 1000}{M}$$

M – molarity of $AgBr$ solution

8 (solubility in g/L) = $\frac{K \times 1000}{(x+y)} \times 188$

42. A

$\left(Ni(CN)_4\right)^{2-}$ is dsp^2 hybridised and diamagnetic

43. D

Equivalents of $KMnO_4 = 5x$

Equivalents of $K_2Cr_2O_7 = 6y$

Equivalents of oxidising agent must be same $5x = 6y$

$x : y = 6 : 5$

44. B

Greater the activation energy of a reaction, greater is the temperature dependence of rate constant of reaction.

45. C

$\frac{1}{\lambda} = RZ^2\left(1 - \frac{1}{4}\right)$ for lyman line

46. C

$Na - 1s^2$	$2s^2$	$2p^6$	$3s^1$
$\boxed{1\downarrow}$	$\boxed{1\downarrow}$	$\boxed{1\downarrow}\boxed{1\downarrow}\boxed{1\downarrow}$	$\boxed{1\downarrow}$
$m_l - 0$	0	$+1\ 0\ -1$	0

47. C

Cl_2 is present simultaneously in two equilibria. Decrease in its concentration shift both equilibria forward.

48. D

Point B represents exactly half neutralisation of acid. At this point concentration acid left and its conjugate base are equal.

49. C

$$\left[Br^-\right] = \frac{4\times10^{-13}}{4\times10^{-4}} = 10^{-9}\,M$$

Mass of $NaBr$ added of $200\,ml = 103\times\frac{10^{-9}}{1000}\times200$

$$= 2.06\times10^{-8}\,g$$

50. C

Greater the charge density, greater is hydration.

51. D

Fluorine is most reactive amongst halides and has highest oxidising power

52. A

$$\left[XeF_3\right]^+\left[SbF_6\right]^-$$

$$\downarrow \qquad \downarrow$$

$$sp^3d \qquad sp^3d^2$$

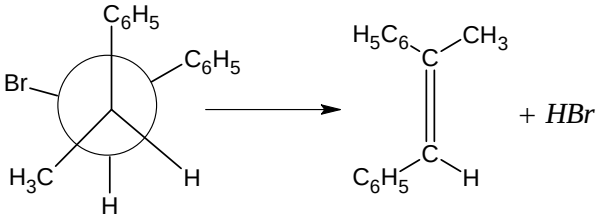
53. A

In $\left(\bullet CF_3\right)$, % p-character because of which bond angle decreases.

54. D

In PCl_2F_3, net dipole moment does not cancel out.

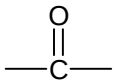
55. A



56. C

$-OH$ group of phenol cannot be replaced by iodine as it does not undergo nucleophilic substitution reaction under normal conditions.

57. A



Group stabilise the carbanion character on $\alpha-$ carbon

58. C

Methoxy group stabilises the cationic intermediate by $+M$ effect

59. B

2 molecules of $RMgX$ are consumed by acid chloride group as it undergoes addition elimination first and then addition. The other aldehydic & ketonic group consume one $RMgX$ molecule each.

60. B

Structure containing a ring of $\left(4n+2\right)\pi e^-s$ is aromatic

Mathematics

PART – III

SECTION – A

61. D

$a=3, b=-4, c=1$

62. D

63. A

$$\int_0^{\frac{\pi}{2}} \frac{(1-\tan x)^2}{(1+\tan x)^4} \sec^2 x\, dx = \frac{1}{3}$$

64. C

Put $x = i$ and equate the real part

65. C

$$x^2 dx + y^2 dy = \frac{d(xy)}{(xy)^2}$$

66. B

Let $x = \frac{\log_2 y}{111}$ these $y^3 - 16y^2 + 8y - 4 = 0$

67. C

$$\sec^{-1}\sqrt{1+x^2} = \tan^{-1} x, \cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1} x$$

68. C

$$\left(\frac{x-2y-4}{\sqrt{5}}\right)^2 = \frac{12}{\sqrt{5}}\left(\frac{2x+y}{\sqrt{5}}\right)$$ (if is parabola)

69. C

$\tan\frac{\theta}{2}\tan\frac{\theta}{2} = \frac{1+e}{1-e}$ if focus is $(-ae, \theta)$

70. B

$$\frac{\lfloor 7}{\lfloor 4 \ \lfloor 2} = 105$$

71. D

$$f(x) = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

72. B

73. B

74. D

75. C

76. B

77. B

$b^2 + c^2 - a^2, c^2 + a^2 - b^2, a^2 + b^2 - c^2$ are in AP.

78. B

$$OM = \sqrt{\left(50\sqrt{5}\right)^2 - (100)^2} = 50m$$

79. B
80. B
81. D

$\Delta = 0$

82. A
83. D

$3+5+7+....\text{upto}\,32\text{nd terms}=1023$

84. C

$$\sin^{-1}\frac{\frac{2}{x}}{1+\left(\frac{1}{x}\right)^2} = 2\tan^{-1}\frac{1}{x}$$

85. D

Both the roots opposite in sign

86. A

$adj\left(AB\right) = adjB.adjA$

87. C

$$\int_1^3 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right)+\tan^{-1}\left(\frac{x^2+1}{x}\right)\right)dx = 2\pi$$

88. C
89. C
90. A

$\vec{a}+\vec{b}=\vec{c}$